

SCI225 Week 8 — Complete Study Guide

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Hematologic Structure & Function · Anemia · Bleeding & Clotting · White Cell Disorders · Pediatric Hematology · Environmental Effects

Study Spine — for every blood disorder ask:

1. Which line is affected (RBC / WBC / platelet)?
2. Production failure or destruction?
3. Compensatory response present?
4. What's the first lab clue?

MECHANISM

WHY

COMPENSATION

PATHOLOGY

MISSTEP

ACRONYM

EXAM TIP

Course alignment

- **LO1** — Recall the structure and function of the hematologic system (Unit 1)
- **LO2** — Describe the alterations of hematologic function and resultant consequences (Units 2–4)
- **LO3** — Describe the alterations of hematologic function in children (Unit 5)
- **LO4** — Connect key ideas from current events to learning (Unit 6)

Unit 1 • Blood Structure & Function LO1

1.1 Erythrocytes (RBCs) and Oxygen Delivery

MECHANISM RBCs are biconcave disks packed with **hemoglobin (Hgb)**, a protein with four globin chains and four heme (iron-containing) groups. Each Hgb molecule carries up to 4 O₂ molecules. Lifespan: ~120 days. Production happens in **red bone marrow** and is driven by **erythropoietin (EPO)** from the kidney.

↓ Tissue O₂ → Kidney senses hypoxia → EPO release → Bone marrow produces more RBCs → ↑ O₂-carrying capacity → Tissue oxygenation restored

Lab	Normal (adult)	What it reflects
Hemoglobin (Hgb)	M: 13.5–17.5 g/dL · F: 12–15.5 g/dL	O ₂ -carrying protein in RBCs
Hematocrit (Hct)	M: 41–53% · F: 36–46%	% of blood volume that is RBCs
RBC count	4.5–5.5 million/μL	Cell number
MCV (mean corpuscular volume)	80–100 fL	Average RBC size — <80 microcytic, >100 macrocytic
Reticulocyte count	0.5–1.5%	Young RBCs — marker of marrow response

WHY Why oxygen delivery is the bottom line. The whole reason to have RBCs is to carry oxygen to tissues. **O₂ delivery = cardiac output × arterial O₂ content.** Any fall in Hgb, SaO₂, or cardiac output reduces delivery. The body responds with *both* long-term increases in RBC production (EPO) and short-term increases in heart rate and respiratory rate to compensate.

1.2 Leukocytes (WBCs)

WBC	Type	Main job
Neutrophil (~60%)	Granulocyte (myeloid)	Phagocytosis — first responder to bacteria; the most direct killer of microorganisms
Lymphocyte (~30%)	Lymphoid	B cells make antibodies; T cells provide cellular immunity (Week 2)
Monocyte / Macrophage (~5%)	Myeloid	Phagocytosis in tissues, antigen presentation
Eosinophil (~3%)	Granulocyte	Parasites, allergic reactions
Basophil (<1%)	Granulocyte	Releases histamine; allergic responses

★ **EXAM TIP** "Which leukocyte function most directly eliminates invading microorganisms?" Phagocytosis — neutrophils and macrophages literally engulf and digest pathogens. Antibody production is more indirect (it tags pathogens for phagocytosis or complement-mediated killing).

1.3 Platelets and Primary Hemostasis

MECHANISM Platelets are anucleate cell fragments shed from megakaryocytes in the marrow. Lifespan ~8–10 days. Normal count: **150,000–400,000/μL**. When a vessel is injured, platelets adhere (via von Willebrand factor) → activate → aggregate → form a temporary **platelet plug** (primary hemostasis). The clotting cascade then deposits fibrin (secondary hemostasis) to stabilize the clot.

Vessel injury → vasoconstriction → platelet adhesion + activation + aggregation → platelet plug → coagulation cascade → fibrin mesh stabilizes clot

Bleeding patterns by deficit type

Deficit	Bleeding pattern
Platelet problem (thrombocytopenia or dysfunction)	Mucocutaneous: petechiae, ecchymoses, gingival bleeding, epistaxis, prolonged bleeding from minor cuts
Coagulation factor problem (e.g., hemophilia)	Deep bleeding: hemarthroses, intramuscular hematomas, delayed bleeding
Vessel problem	Easy bruising, palpable purpura

Practice Questions — Blood Structure & Function

Q1. Which leukocyte function is most directly responsible for eliminating invading microorganisms?

- A. Antibody production by plasma cells
- B. Phagocytosis by neutrophils and macrophages
- C. Histamine release by basophils
- D. Cytokine secretion by helper T cells

[Show answer](#)

B. Phagocytes (neutrophils first, macrophages in tissue) directly engulf and digest pathogens. Antibodies tag, but the killing is largely done by phagocytes and complement.

Q2. Which finding would the nurse most expect in a patient with thrombocytopenia?

- A. Hemarthroses (deep joint bleeds)
- B. Petechiae and easy bruising
- C. Pallor and fatigue only
- D. Recurrent bacterial pneumonia

[Show answer](#)

B. Platelet deficiency produces mucocutaneous bleeding (petechiae, ecchymoses, mucosal bleeding). Hemarthroses suggest a coagulation factor problem (e.g., hemophilia).

Q3. After blood donation, which mechanism most directly restores red blood cell count over the following weeks?

- A. Splenic release of stored RBCs
- B. Erythropoietin from the kidneys stimulates marrow erythropoiesis
- C. Hepatic synthesis of new hemoglobin in the liver
- D. Lymphocyte proliferation

[Show answer](#)

B. Reduced O₂-carrying capacity is sensed by the kidneys, which release EPO → bone marrow ramps up RBC production → reticulocyte count rises → Hgb returns toward baseline over weeks.

Unit 2 • Anemia LO2

MECHANISM Anemia = decreased Hgb / RBCs / Hct → reduced O₂-carrying capacity → tissue hypoxia. The body compensates with several mechanisms; only severe or rapidly developing anemia overwhelms compensation.

2.1 Major Anemia Types — Group by Mechanism

Anemia	MCV	Mechanism	Clinical clue
Iron deficiency	Microcytic (low MCV), hypochromic	Insufficient iron → can't make heme → small, pale RBCs	Chronic blood loss (GI bleed, menstrual), poor diet, pregnancy; pica; spoon nails

Vitamin B12 deficiency	Macrocytic (high MCV)	B12 needed for DNA synthesis; deficiency → impaired RBC division → large immature cells; pernicious anemia = autoimmune loss of intrinsic factor → can't absorb B12	Glossitis, paresthesias, ataxia (B12 also needed for myelin)
Folate deficiency	Macrocytic	Same DNA synthesis problem as B12; from poor diet, alcoholism, pregnancy	No neurologic findings (key difference from B12)
Hemolytic anemia	Often normocytic	RBCs destroyed faster than made → free Hgb → indirect bilirubin → jaundice; ↑ reticulocytes	Anemia + jaundice + ↑ indirect bilirubin + ↑ LDH; sickle cell, G6PD, autoimmune, mechanical valve
Anemia of chronic disease (inflammation)	Normocytic or microcytic	Chronic inflammation → hepcidin ↑ → iron locked in storage despite adequate stores → marrow can't access it	Chronic infection, autoimmune disease, malignancy, CKD; ferritin normal or HIGH; develops after viral inflammation in some patients
Aplastic anemia	Normocytic	Marrow failure → pancytopenia (low RBCs, WBCs, platelets)	Radiation, drugs, chemicals, viral; bleeding + infection + fatigue
Acute blood loss	Normocytic initially	Hemorrhage; Hgb often normal at first (hemoconcentrated) until volume replaced	Trauma, GI bleed; tachycardia, hypotension

ACRONYM / MNEMONIC "Microcytic = small Iron / chronic disease. Macrocytic = big B12 / Folate."

Reading the MCV is your first move on any anemia question — it splits the differential immediately.

PATHOLOGY Hemolytic anemia signature. When RBCs are destroyed, the body sees: anemia + jaundice (from heme breakdown to bilirubin) + elevated indirect bilirubin + elevated LDH (released from lysed cells) + low haptoglobin (binds free Hgb). The reticulocyte count is typically high because the marrow tries to keep up.

WHY Why anemia of chronic disease has paradoxically adequate iron stores. Inflammatory cytokines (IL-6) raise **hepcidin**, a hormone that locks iron inside macrophages and blocks intestinal absorption. Total body iron is fine — it's just unavailable to the marrow. Iron supplementation alone won't fix it; treating the underlying disease will.

2.2 Compensatory Responses to Anemia

COMPENSATION Cardiovascular ↑ heart rate, ↑ stroke	COMPENSATION Respiratory ↑ respiratory rate and	COMPENSATION Marrow ↑ EPO from kidneys → ↑
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volume, ↑ cardiac output
→ push more blood per
minute through tissues

depth → maximize SaO₂
and O₂ uptake

erythropoiesis → ↑
reticulocyte count

MECHANISM At the tissue level, oxygen unloading from Hgb increases (right shift of the O₂-Hgb dissociation curve) thanks to ↑ 2,3-BPG, ↑ CO₂, ↓ pH. This means each Hgb molecule gives up more of its O₂ to tissues that need it most.

★ **EXAM TIP** "Which mechanisms directly improve oxygen availability at the tissue level?" Answers include: increased cardiac output, increased respiratory rate, increased EPO-driven erythropoiesis, and right shift of the O₂-Hgb dissociation curve (more O₂ released to tissues).

Practice Questions — Anemia

Q4. A patient has microcytic, hypochromic erythrocytes. Most likely cause?

- A. Vitamin B12 deficiency
- B. Folate deficiency
- C. Iron deficiency
- D. Acute blood loss

[Show answer](#)

C. Microcytic + hypochromic = iron deficiency. Without iron, heme synthesis fails and RBCs are smaller and paler.

Q5. A patient with chronic inflammation develops anemia despite adequate iron stores. Which mechanism is most responsible?

- A. Loss of intrinsic factor
- B. Hepcidin elevation locking iron in macrophages and blocking absorption
- C. Bone marrow infiltration
- D. Hemolysis from antibody binding

[Show answer](#)

B. Anemia of chronic disease — IL-6 → hepcidin → iron stays sequestered. Total body iron is normal/high but functionally unavailable. Treating inflammation is the answer.

Q6. A patient presents with anemia, jaundice, and elevated indirect bilirubin. Which type of anemia is most likely?

- A. Iron deficiency
- B. Hemolytic anemia
- C. Folate deficiency
- D. Acute blood loss

[Show answer](#)

B. Hemolysis releases heme → broken down to bilirubin → unconjugated (indirect) bilirubin rises → jaundice. ↑ LDH and ↑ reticulocytes complete the picture.

Q7. A patient is diagnosed with vitamin B12 deficiency anemia. Which finding is most expected?

- A. Microcytic, hypochromic RBCs
- B. Macrocytic RBCs and possible neurologic symptoms (paresthesia, ataxia)
- C. ↑ indirect bilirubin and jaundice
- D. Pancytopenia

[Show answer](#)

B. B12 deficiency → impaired DNA synthesis → macrocytic RBCs. B12 is also required for myelin synthesis → neurologic symptoms differentiate it from folate deficiency.

Q8. Select all that apply. Which mechanisms directly improve oxygen availability at the tissue level in anemia?

- A. Increased cardiac output
- B. Increased respiratory rate and depth
- C. Increased erythropoietin → erythropoiesis
- D. Right shift of the O₂-Hgb dissociation curve
- E. Reduced kidney function

[Show answer](#)

A, B, C, D. All four are physiologic compensations for hypoxia. E is the opposite — kidney function (and EPO) declining would worsen anemia.

Q9. A patient with chronic hypoxia (e.g., chronic lung disease) develops increased red blood cell production. Which best explains this?

- A. The bone marrow randomly increases output
- B. Hypoxia → ↑ EPO from the kidneys → marrow erythropoiesis
- C. Inflammation drives more erythrocytes
- D. Pancreatic hormones rise

[Show answer](#)

B. Same axis as in altitude adaptation — kidneys sense hypoxia and produce EPO, which drives marrow to make more RBCs. Hct can rise enough to increase blood viscosity and thrombosis risk.

Unit 3 • Bleeding & Clotting Disorders LO2

3.1 Thrombocytopenia

MECHANISM Thrombocytopenia = platelet count < 150,000/μL. Bleeding risk rises as count falls. Spontaneous bleeding is rare above 50,000; serious risk < 20,000; life-threatening < 10,000.

Mechanisms:

- **Decreased production** — marrow failure (aplastic anemia, leukemia, chemotherapy, radiation, B12 deficiency)
- **Increased destruction** — immune (ITP), drug-induced, DIC, TTP/HUS, hypersplenism
- **Sequestration** — splenic enlargement traps platelets

Classic presentation: petechiae, ecchymoses, mucosal bleeding (gums, nose), prolonged bleeding from minor cuts. Heavy menstruation. No deep bleeds (those suggest coagulation factor problems).

3.2 Polycythemia and Thrombosis

PATHOLOGY Polycythemia = elevated RBC mass (Hct typically > 50%). Causes:

- **Primary (polycythemia vera)** — myeloproliferative disorder; bone marrow makes too many RBCs autonomously
- **Secondary appropriate** — chronic hypoxia (high altitude, COPD, smoking, sleep apnea, congenital heart disease) → ↑ EPO → ↑ RBC
- **Secondary inappropriate** — EPO-secreting tumor (renal cell carcinoma)

WHY Why polycythemia raises thrombosis risk. Increased Hct = increased blood viscosity. Slower flow + more cells touching each other = greater platelet aggregation and clot formation. Patients can develop strokes, MIs, DVT, PE. Treatment includes phlebotomy and management of the underlying cause.

3.3 Thrombosis — Virchow's Triad

ACRONYM / MNEMONIC Virchow's triad — three drivers of thrombosis:

1. **Endothelial injury** — atherosclerosis, surgery, trauma, central line
2. **Stasis (slow flow)** — immobility, atrial fibrillation, varicose veins, prolonged sitting
3. **Hypercoagulability** — inherited (Factor V Leiden, antithrombin deficiency), acquired (cancer, pregnancy, oral contraceptives, polycythemia)

PATHOLOGY Immobility-related thrombosis hits all three Virchow drivers in a hospitalized or post-surgical patient: stasis from lying still, endothelial injury from surgery, hypercoagulable state from inflammation. Prophylactic anticoagulation, sequential compression devices, and early ambulation are all preventive.

Practice Questions — Bleeding & Clotting

Q10. A patient with thrombocytopenia develops prolonged bleeding after minor injury. Which mechanism best explains the prolonged bleeding?

- A. Coagulation factor deficiency
- B. Inadequate platelet plug formation due to low platelet count
- C. Increased fibrinolysis
- D. Endothelial cell dysfunction

Show answer

B. Platelets are the first responders to vessel injury, forming the primary hemostatic plug. With too few platelets, the plug doesn't form effectively and minor injuries bleed longer than expected.

Q11. A patient with polycythemia develops increased risk of thrombosis. The best explanation is:

- A. Excess platelet count alone
- B. Increased blood viscosity from elevated RBC mass slows flow and promotes clot formation
- C. Hypoxia-induced vasodilation
- D. Iron deficiency

Show answer

B. More RBCs = thicker blood = slower flow + more platelet aggregation. Strokes, MIs, and DVTs are all more common in untreated polycythemia.

Q12. A patient who has been immobile for several days develops thrombosis. Which Virchow's triad component most directly explains this?

- A. Endothelial injury alone
- B. Venous stasis from prolonged immobility
- C. Decreased hepatic clotting factor synthesis
- D. Decreased platelet count

Show answer

B. Stasis is the dominant driver in immobile patients (though inflammation/hypercoagulability also play in). Early mobilization, compression devices, and prophylactic anticoagulation prevent this.

Unit 4 • White Cell Disorders LO2

4.1 Leukopenia and Neutropenia

MECHANISM **Leukopenia** = WBC count < 4,000/ μ L. **Neutropenia** = absolute neutrophil count (ANC) < 1,500. Severe neutropenia (ANC < 500) is a medical emergency for infection risk.

Common causes:

- Chemotherapy — most common cause; nadir typically 7–14 days after dose
- Radiation
- Bone marrow infiltration (leukemia, lymphoma, metastases)
- Aplastic anemia
- Severe infections (paradoxically — sepsis can deplete neutrophils)
- Autoimmune neutropenia
- Drug-induced (clozapine, methimazole, others)
- B12 / folate deficiency

PATHOLOGY The **neutropenic patient** can have profound infection without classic signs because they cannot mount inflammation. Fever may be the only sign. **Neutropenic fever** requires immediate broad-spectrum IV antibiotics — even before culture results.

4.2 Leukemia (Brief Recap from Week 3)

Type	Age	Pattern
ALL (acute lymphoblastic)	Children 2–5	Most common pediatric cancer; lymphoblasts crowd marrow
AML (acute myeloid)	Adults	Myeloblasts; Auer rods on smear
CML (chronic myeloid)	Middle-age adults	Philadelphia chromosome t(9;22), BCR-ABL fusion
	Older adults	Often asymptomatic; smudge cells

CLL (chronic lymphocytic)		
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WHY Why leukemia patients get infections. Even though the WBC count may look high on labs, those cells are *blasts* — immature, nonfunctional. The patient functionally has leukopenia. Marrow infiltration also crowds out normal hematopoiesis → anemia (fatigue, pallor) + thrombocytopenia (petechiae) + leukopenia (infections) all together.

Practice Questions — White Cell Disorders

Q13. A patient with leukopenia develops frequent infections. Which best explains this?

- A. Increased platelet aggregation
- B. Decreased numbers of functional white blood cells (especially neutrophils) impair pathogen clearance
- C. Reduced erythropoietin production
- D. Enhanced antibody response

[Show answer](#)

B. Without enough neutrophils and other WBCs, the immune system can't clear bacteria, fungi, and other pathogens efficiently. Neutropenic fever requires immediate empirical antibiotics.

Q14. A patient receiving chemotherapy develops a low white blood cell count. Best explanation?

- A. Chemotherapy specifically targets only WBCs
- B. Chemotherapy is cytotoxic to rapidly dividing cells, including marrow precursors → decreased WBC production
- C. Cytokine release directly destroys mature neutrophils
- D. Reduced kidney EPO production

[Show answer](#)

B. Chemo doesn't discriminate well — it hits all rapidly dividing cells, including bone marrow precursors. Result: predictable cytopenias 7–14 days after dosing. Neutrophil nadir is the highest infection-risk window.

Q15. A child with leukemia develops frequent infections. Which mechanism best explains this?

- A. The leukemic cells are functional and protect against infection
- B. Marrow infiltration with leukemic blasts crowds out normal WBC production, and blasts themselves are nonfunctional
- C. The patient is hypercoagulable
- D. The patient is iron deficient

[Show answer](#)

B. Even with high "WBC counts" on labs, leukemic blasts can't fight infection, and the marrow can't make functional cells.

Unit 5 • Pediatric Hematology LO3

5.1 Sickle Cell Disease

MECHANISM Autosomal recessive mutation in the β -globin gene → abnormal **hemoglobin S (HbS)**. Under low O₂, acidosis, dehydration, or stress, HbS polymerizes, deforming RBCs into rigid, sickle-shaped cells. These cells:

- Block small vessels → **vaso-occlusive crises** → pain, organ ischemia
- Are destroyed faster than normal → **chronic hemolytic anemia**
- Damage the spleen by repeated infarctions → **functional asplenia** → susceptibility to encapsulated bacteria (*S. pneumoniae*, *H. influenzae*, *Salmonella*)

Classic features:

- Recurrent severe pain crises (back, chest, extremities, abdomen)
- Acute chest syndrome — sickling in pulmonary vessels (medical emergency)
- Splenic sequestration (acute drop in Hgb in young children)
- Stroke (pediatric stroke risk significantly elevated)
- Dactylitis (hand-foot swelling from infarction in small bones)
- Chronic hemolytic anemia with elevated reticulocyte count
- Increased gallstones (chronic bilirubin overload)

★ **EXAM TIP Triggers of vaso-occlusive crisis:** dehydration, hypoxia, acidosis, infection, cold exposure, stress. Treat acutely with hydration, oxygen, pain control. Prevent with hydroxyurea (raises HbF, which doesn't sickle).

5.2 Neonatal Jaundice and Hemolytic Disease of the Newborn

MECHANISM Physiologic jaundice appears **after** 24 hours of life and resolves within a week. It reflects the immature liver's limited ability to conjugate the bilirubin produced as fetal hemoglobin is broken down.

Pathologic jaundice appears **within 24 hours** of birth and signals abnormal hemolysis or other pathology. Unconjugated bilirubin can cross the BBB and cause **kernicterus** — irreversible neurologic damage.

PATHOLOGY Hemolytic disease of the newborn (HDN). Rh-negative mother previously sensitized to Rh-positive fetal blood (often during a prior pregnancy or delivery) makes IgG anti-Rh antibodies. In a subsequent Rh-positive pregnancy, these antibodies cross the placenta and destroy fetal RBCs → fetal anemia, jaundice, hydrops fetalis. **RhoGAM** (anti-D Ig given at 28 weeks and within 72 h of delivery) prevents maternal sensitization.

WHY Why a newborn with jaundice within 24 hours is a red flag. Physiologic jaundice can't appear that fast — it requires the neonatal liver to gradually fall behind. Within-24-hour jaundice means active hemolysis (HDN, ABO incompatibility, G6PD deficiency, sepsis). Don't wait.

Premature infant anemia

MECHANISM Premature infants commonly develop anemia during their first weeks of life. Causes: shortened RBC lifespan, frequent blood draws (a substantial portion of small

infants' blood volume), low EPO production by the immature kidney, rapid growth diluting Hgb. Often requires transfusion or supplemental EPO.

Practice Questions — Pediatric Hematology

Q16. A child with sickle cell disease develops recurrent painful crises. Which mechanism best explains the pain?

- A. Increased platelet count blocks vessels
- B. Sickled RBCs occlude small blood vessels, causing tissue ischemia and pain
- C. Increased RBC production overworks the marrow
- D. Iron deficiency

[Show answer](#)

B. Vaso-occlusive crisis. Sickled cells are rigid and block microvasculature → ischemic pain. Triggers include dehydration, hypoxia, infection, cold, and stress. Hydration, O₂, pain control are mainstays.

Q17. A newborn develops jaundice within 24 hours of birth. Which mechanism best explains this?

- A. Normal physiologic jaundice from immature liver
- B. Pathologic hemolysis (e.g., HDN, ABO incompatibility, G6PD deficiency)
- C. Healthy weight gain
- D. Brown adipose tissue activation

[Show answer](#)

B. Within 24 hours = pathologic. Active hemolysis is the most common cause; check Rh and ABO status, hemolysis labs. Bilirubin crossing the BBB causes kernicterus.

Q18. A newborn with hemolytic disease of the newborn develops anemia. The condition is best prevented by:

- A. Routine vitamin K injection
- B. RhoGAM (anti-D immunoglobulin) given to Rh-negative mothers at 28 weeks and after delivery
- C. Folic acid in the third trimester
- D. Iron supplementation in pregnancy

[Show answer](#)

B. RhoGAM binds and removes any fetal Rh-positive RBCs that have entered maternal circulation, preventing the mother from making lasting anti-Rh antibodies.

Q19. A child with anemia develops increased heart rate and respiratory rate. Which mechanism best explains these compensatory changes?

- A. Loss of vagal tone unrelated to anemia
- B. The body increases cardiac output and respiratory rate to compensate for reduced oxygen delivery
- C. Iron toxicity
- D. Vitamin overdose

[Show answer](#)

B. Anemia → reduced O₂ delivery → reflex sympathetic activation → ↑ HR + ↑ RR + ↑ stroke volume to maintain tissue oxygenation.

Q20. A premature infant develops anemia during the first weeks of life. Which factor most directly contributes?

- A. Excessive iron intake
- B. Low neonatal EPO production by the immature kidney plus shortened RBC lifespan and frequent blood draws
- C. Increased erythropoiesis
- D. Early gut microbiome

[Show answer](#)

B. "Anemia of prematurity" — multifactorial: low EPO, fragile RBCs, blood loss from sampling, rapid growth diluting Hgb. May need transfusion or recombinant EPO.

Q21. A child with chronic anemia develops an increased respiratory rate. Which mechanism best explains this change?

- A. Acidosis from kidney failure
- B. Compensatory increase in ventilation to maximize oxygen uptake and delivery
- C. Hyperventilation from anxiety only
- D. Pulmonary embolism

[Show answer](#)

B. Chronic anemia → low tissue O₂ → chemoreceptor and central drive to breathe more, raising SaO₂ and total O₂ uptake.

Q22. A child with anemia shows a compensatory rise in reticulocyte count. This finding indicates:

- A. The marrow is failing
- B. The marrow is responding appropriately by ramping up RBC production
- C. Hemolysis has been suppressed
- D. Iron deficiency is ruled out

[Show answer](#)

B. Reticulocytes are immature RBCs newly released from marrow. A high count means the marrow recognizes the anemia and is responding. A low count in anemia means the marrow is failing.

Unit 6 • Environmental & Systemic Effects LO4

6.1 High Altitude — Adaptive Polycythemia

MECHANISM At high altitude, atmospheric O₂ partial pressure is lower → lower SaO₂ → tissue hypoxia. The body adapts:

1. Acute: hyperventilation increases SaO₂ partially
2. Days to weeks: **kidney releases EPO** → bone marrow produces more RBCs → Hct rises

3. Right shift of O₂-Hgb dissociation curve (↑ 2,3-BPG) → more O₂ unloaded to tissues
This is "secondary appropriate" polycythemia. It improves O₂ delivery but raises blood viscosity and thrombosis risk.

6.2 Radiation Exposure

PATHOLOGY Ionizing radiation damages rapidly dividing bone marrow precursors. Severe exposure → **aplastic anemia / pancytopenia**: low RBCs (anemia), low platelets (bleeding), low WBCs (infection). The marrow is hypocellular on biopsy. Treatment: supportive care, transfusions, growth factors, possible stem-cell transplant.

6.3 Environmental Toxins

Several environmental exposures damage the marrow or RBCs directly:

- **Benzene** — aplastic anemia, leukemia (refining, gasoline, plastics)
- **Lead** — microcytic anemia (interferes with heme synthesis), basophilic stippling, neurotoxicity in children
- **Radiation** — pancytopenia, leukemia
- **Pesticides, solvents** — variable marrow effects

6.4 Viral Infection and Anemia

MECHANISM Some viral infections trigger systemic inflammation that decreases hemoglobin via the same hepcidin-mediated mechanism as anemia of chronic disease. Inflammatory cytokines suppress erythropoiesis and limit iron availability. The anemia is typically mild, normocytic, and resolves with the infection. Severe viral illness (parvovirus B19) can directly suppress RBC precursors, causing transient aplastic crisis — particularly dangerous in patients with chronic hemolysis like sickle cell disease.

6.5 Post-Donation Recovery

After whole-blood donation (~500 mL):

- Plasma volume restores in 24–48 h via fluid intake and shift from interstitium
- RBC mass recovers over 4–8 weeks via EPO-driven marrow erythropoiesis
- Iron stores can take longer; iron supplementation often recommended for frequent donors

Practice Questions — Environmental & Systemic

Q23. Individuals living at high altitude develop hematologic changes. Which mechanism best explains this adaptation?

- A. Hypoxia stimulates renal EPO release, driving increased RBC production
- B. Cold air increases bone marrow temperature
- C. UV radiation directly produces RBCs
- D. Lower atmospheric pressure compresses blood vessels

Show answer

A. Same kidney-EPO-marrow axis seen in chronic hypoxia patients. Hct can rise significantly with prolonged altitude exposure.

Q24. A patient exposed to radiation develops pancytopenia. Which best explains this?

- A. Radiation directly destroys mature RBCs in the circulation
- B. Ionizing radiation damages rapidly dividing bone marrow precursors, reducing all blood cell lines
- C. Radiation activates platelets to consume themselves
- D. Radiation only affects red blood cells

[Show answer](#)

B. Marrow precursors are highly proliferative and very radiation-sensitive. Loss of stem and precursor cells → pancytopenia. Treatment requires marrow recovery, transfusions, and possibly stem cell transplant.

Q25. A patient develops anemia following a viral infection that triggers systemic inflammation and decreases hemoglobin. The most likely mechanism is:

- A. Hepcidin elevation from inflammation locks iron in stores and suppresses erythropoiesis
- B. Vitamin K deficiency
- C. Acute hemolysis from antibody binding
- D. Bone marrow tumor infiltration

[Show answer](#)

A. Inflammation → IL-6 → hepcidin → iron sequestration + impaired erythropoiesis. Same anemia-of-chronic-disease mechanism applied to acute viral illness.

Q26. A population exposed to environmental toxins develops anemia. Which exposure most directly causes microcytic anemia in children by interfering with heme synthesis?

- A. Benzene
- B. Lead
- C. Aluminum cookware
- D. Carbon dioxide

[Show answer](#)

B. Lead inhibits enzymes in heme synthesis (ALA dehydratase, ferrochelatase) → microcytic anemia with basophilic stippling. Children are especially vulnerable to neurotoxicity from environmental lead exposure.

Appendix • Top Missteps & Master Mnemonics

Top 15 Week 8 Missteps

#	The trap	What's actually true
1	"Microcytic anemia is B12 deficiency."	Microcytic = iron deficiency (or chronic disease, thalassemia). B12/folate = MACROcytic.
2	"Anemia of chronic disease has low iron stores."	Iron stores are NORMAL or HIGH. The problem is iron is locked in macrophages by hepcidin.
3		

	"Hemolytic anemia has low reticulocytes."	Hemolysis triggers HIGH reticulocytes (marrow trying to keep up).
4	"B12 deficiency has no neurologic symptoms."	B12 is needed for myelin — paresthesias, ataxia. This is what distinguishes it from folate deficiency.
5	"Thrombocytopenia causes deep joint bleeds."	Platelet problems = mucocutaneous (petechiae, gum bleeding). Hemarthroses = coagulation factor problems.
6	"Polycythemia helps oxygen delivery without risk."	↑ Hct → ↑ viscosity → thrombosis risk (stroke, MI, DVT).
7	"Newborn jaundice is always normal."	Within 24 hours = pathologic (hemolysis). Physiologic jaundice appears AFTER 24 hours.
8	"RhoGAM is given to Rh-positive mothers."	RhoGAM is for Rh-negative mothers carrying potentially Rh-positive fetuses.
9	"Sickle cell crisis is treated with diuretics."	Treat with HYDRATION (sickling worsens with dehydration), oxygen, and pain control.
10	"Leukemia patients have functional WBCs."	Blasts are nonfunctional even when count looks high. Patients have functional leukopenia.
11	"Chemotherapy WBC drop is unrelated to infection risk."	It IS the infection risk. Neutropenic fever requires immediate empirical antibiotics.
12	"Thrombosis after immobility is rare."	Stasis is one of three Virchow drivers; immobility is a major DVT risk.
13	"EPO comes from the bone marrow."	EPO comes from the kidneys (responding to hypoxia). Marrow is the target of EPO.
14	"Phagocytes only eat dead cells."	Neutrophils and macrophages actively eat live invading bacteria — primary microbial killing.
15	"Lead anemia is macrocytic."	Lead causes microcytic anemia by inhibiting heme synthesis enzymes; basophilic stippling on smear.

Master Mnemonic Sheet

Mnemonic	Meaning
Micro = Iron / Chronic / Lead	Low MCV anemia categories
Macro = B12 / Folate	High MCV anemia categories
"B12 has neuro, folate doesn't"	Key differentiator between the two macrocytic anemias
Hemolysis triad	↑ indirect bilirubin + ↑ LDH + ↓ haptoglobin + ↑ retics
Virchow's triad	Endothelial injury + Stasis + Hypercoagulability
"S = Soaked, D = Dry"	From Week 6 — applies again to fluid-driven anemia decisions
Within 24 h jaundice = pathologic	Investigate hemolysis

RhoGAM = Rh-NEG mom	Given at 28 weeks and within 72 h of delivery
Sickle triggers = "Cold, Hypoxia, Acidosis, Dehydration, Stress, Infection"	Avoidance prevents crises
Neutropenic fever = empirical antibiotics	Don't wait for cultures
EPO from kidneys	Hypoxia sensor → marrow erythropoiesis driver
Hepcidin = iron lockup	The hormone of anemia of chronic disease

Anemia Differential Quick Reference

Pattern	Most likely
Microcytic + chronic blood loss	Iron deficiency
Microcytic + chronic disease, normal/high ferritin	Anemia of chronic disease
Microcytic + child with environmental exposure	Lead poisoning
Macrocytic + glossitis + paresthesia	B12 deficiency
Macrocytic + alcoholism or pregnancy	Folate deficiency
Anemia + jaundice + ↑ indirect bilirubin + ↑ LDH	Hemolytic anemia
Pancytopenia after radiation/chemo/benzene	Aplastic anemia
Polycythemia at altitude or with COPD	Secondary appropriate (EPO-driven)
Newborn with jaundice in first 24 h	Hemolytic disease (HDN, ABO, G6PD)
Child with pain crises and chronic anemia	Sickle cell disease

Week 8 Study Strategy

- **Read MCV first on every anemia question.** It splits the differential before you read another lab.
- **Pair every anemia type with its mechanism.** Iron = no heme. B12 = no DNA + no myelin. ACD = hepcidin lockup. Hemolytic = destruction. Aplastic = marrow gone.
- **Memorize the compensations cold.** ↑ HR, ↑ RR, ↑ EPO, right shift O₂-Hgb curve. SATA classics.
- **Don't mix bleeding patterns.** Mucocutaneous = platelet. Deep / hemarthroses = factor.
- **"Within 24 hours" jaundice in newborns is pathologic.** Don't pick the "physiologic" distractor.
- **Sickle cell triggers** are extremely high yield — dehydration, hypoxia, acidosis, infection, cold, stress.
- **Connect environmental exposures to mechanisms.** Altitude = EPO. Radiation = pancytopenia. Lead = microcytic. Benzene = aplastic / leukemia.

SCI225 Week 8 Study Guide — built from the Week 8 learning objectives covering hematologic structure and function, anemias, bleeding and clotting disorders, white cell disorders, pediatric hematology, and environmental influences. Made by Dr. Vidya Aggrawal to teach the concepts tested, not the exam questions themselves. Same color-coded format as Weeks 1–6. Print for paper, or use in-browser with collapsible practice answers.